

# Yuqian Zhang

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## Positions

**Assistant Professor**

September 2022 – present

Institute of Statistics and Big Data, Renmin University of China

## Education

**University of California, San Diego, U.S.**

September 2018 – March 2022

Ph.D. in Mathematics with a Specialization in Statistics

Advisor: Jelena Bradic

**University of California, San Diego, U.S.**

September 2016 – June 2018

M.S. in Statistics

**Wuhan University, China**

September 2012 – June 2016

B.S. in Mathematics and Applied Mathematics

## Research Interests

- Causal Inference
- Semi-supervised Learning
- Dynamic Treatment Regimes
- Machine Learning and Reinforcement Learning
- High-dimensional Statistics

## Publications (\* corresponding, # alphabetical order, † equal contribution, \_\_ students)

**Yuqian Zhang**<sup>†</sup>, Abhishek Chakraborty<sup>†</sup>, and Jelena Bradic<sup>†</sup> (2026). The decaying missing-at-random framework: doubly robust causal inference with partially labeled data. *IEEE Transactions on Information Theory*, accepted. arXiv:2305.12789.

Kai Chen and **Yuqian Zhang**<sup>\*</sup> (2025). Semi-supervised linear regression: enhancing efficiency and robustness in high dimensions. *Biometrics*, 81(3), uja113. (*Kai received the Second Prize for Outstanding Papers at the 8th National Statistics Doctoral Academic Forum.*)

Jelena Bradic<sup>#</sup>, Weijie Ji<sup>#</sup>, and **Yuqian Zhang**<sup>#</sup> (2024). High-dimensional inference for dynamic treatment effects. *The Annals of Statistics*, 52(2), 415–440.

**Yuqian Zhang**<sup>†</sup>, Abhishek Chakraborty<sup>†</sup>, and Jelena Bradic<sup>†</sup> (2023). Double robust semi-supervised inference for the mean: selection bias under MAR labeling with decaying overlap. *Information and Inference: A Journal of the IMA*, 12(3), 2066–2159.

**Yuqian Zhang** and Jelena Bradic (2022). High-dimensional semi-supervised learning: in search of optimal inference of the mean. *Biometrika*, 109(2), 387–403. (*This paper received the 2021 Student Paper Award from the ASA Nonparametric Statistics Section.*)

Zheng Zhang, Kun Zhang, Xing Yan, Songshan Yang, and **Yuqian Zhang** (2026). *Big Data in*

## Preprints

Zhen Qi and **Yuqian Zhang**\* (2026). Quadruply robust methods for causal mediation analysis. arXiv:2601.22592. Major revision at *Journal of the American Statistical Association*.

Yilizhati Anniwaer and **Yuqian Zhang**\* (2025). Semi-supervised inference for treatment heterogeneity. arXiv:2509.05048. Major revision at *Biometrics*. (*Yilizhati received the Third Prize for Outstanding Papers at the 9th National Statistics Doctoral Academic Forum.*)

Kai Chen and **Yuqian Zhang**\* (2025). Balancing utility and cost in dynamic treatment regimes. arXiv:2507.17360. Reject and resubmit at *Journal of the Royal Statistical Society: Series B*.

Facheng Yu, Zhen Qi, and **Yuqian Zhang**\* (2024). Semi-supervised inference using unlabeled summary statistics. arXiv:2411.15691.

**Yuqian Zhang** and Jelena Bradic (2024). Causal inference through multi-stage learning and doubly robust deep neural networks. arXiv:2407.08560.

**Yuqian Zhang**, Weijie Ji, and Jelena Bradic (2024). Adaptive split balancing for optimal random forest. arXiv:2402.11228.

**Yuqian Zhang**, Weijie Ji, and Jelena Bradic (2021). Dynamic treatment effects: high-dimensional inference under model misspecification. arXiv:2111.06818. Major revision at *Journal of the Royal Statistical Society: Series B*. (*This paper received the ICOSA 2025 China Conference Junior Researcher Award.*)

## Awards and Honors

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|--------------------------------------------------------------------------------------------------|-------------|
| Junior Researcher Award, International Chinese Statistical Association                           | June 2025   |
| Best Student Paper Award (Nonparametric Statistics Section),<br>American Statistical Association | August 2021 |
| Teaching Assistant Award, Department of Mathematics, UC San Diego                                | 2019 – 2020 |
| Friends of the International Center Scholarship, UC San Diego                                    | May 2019    |
| Outstanding Graduate, Wuhan University                                                           | June 2016   |

## Research Grants

The RUC Endurance Research Fund for Long-Term Basic Studies. “Statistical Foundations of Artificial Intelligence”. Grant No. 26XNKJ31. 2026.07–2031.06. Principal Investigator.

The RUC Start-up Fund. “Semi-supervised Inference under Selection Bias”. Grant No. 23XNKJ18. 2023.01–2025.12. Principal Investigator.

The NSFC Young Scientists Fund. “Statistical Inference Based on High-dimensional Semi-supervised Data”. Grant No. 12301390. 2024.01–2026.12. Principal Investigator.

The NSFC General Program. “Nonparametric Modeling and Inference for Continuous Treatment Effect”. Grant No. 12371284. 2024.01–2027.12. Co-investigator (with Zheng Zhang).

## Presentations

|                                                                                         |           |
|-----------------------------------------------------------------------------------------|-----------|
| The 4th Joint Conference on Statistics and Data Science & IMS-China                     | July 2026 |
| Young Researchers Workshop on Statistics and Data Science                               | July 2026 |
| Department of Mathematics, Hong Kong Baptist University                                 | June 2026 |
| IMS Asia Pacific Rim Meeting 2026                                                       | June 2026 |
| 2026 Annual Meeting of the CAAS Committee on Multivariate Analysis Application          | June 2026 |
| 2026 International Conference on Frontiers of Data Science                              | May 2026  |
| School of Economics and Management, Tsinghua University                                 | Mar 2026  |
| Sanya Workshop on Causality and Machine Learning                                        | Jan 2026  |
| 2025 Xiamen University Symposium on Modern Statistics                                   | Dec 2025  |
| School of Mathematical Sciences, Shanghai Jiao Tong University                          | Dec 2025  |
| 2025 International Conference on Data Science and Brain-Inspired Intelligence           | Dec 2025  |
| School of Statistics & Institute of Statistics and Big Data, Renmin University of China | Sep 2025  |
| The 8th International Conference on Econometrics and Statistics                         | Aug 2025  |
| The 3rd Chinese Causal Inference Conference                                             | July 2025 |
| ICSA 2025 China Conference                                                              | June 2025 |
| School of Mathematics and Statistics, Wuhan University                                  | June 2025 |
| 2025 Annual Meeting of the CAAS Committee on Multivariate Analysis Application          | June 2025 |
| School of Management, University of Science and Technology of China                     | June 2025 |
| School of Statistics, Capital University of Economics and Business                      | May 2025  |
| School of Statistics & Institute of Statistics and Big Data, Renmin University of China | Apr 2025  |
| School of Statistics, East China Normal University                                      | Apr 2025  |
| School of Mathematical Sciences, Tongji University                                      | Apr 2025  |
| School of Management, Fudan University                                                  | Apr 2025  |
| School of Statistics and Mathematics, Central University of Finance and Economics       | Mar 2025  |
| Guanghua School of Management, Peking University                                        | Feb 2025  |
| School of Mathematics and Statistics, Wuhan University                                  | Jan 2025  |
| Causal Inference: Bridging Theory and Practice                                          | Jan 2025  |
| The 18th International Joint Conference CFE-CMStatistics                                | Dec 2024  |
| The 1st Macau International Conference on Business Intelligence and Analytics           | Dec 2024  |
| School of Statistics and Data Science, Nankai University                                | Dec 2024  |
| School of Statistics and Management, Shanghai University of Finance and Economics       | Nov 2024  |
| The 2nd International Conference on Machine Learning and Statistics                     | Aug 2024  |
| The 1st Joint Conference on Statistics and Data Science                                 | July 2024 |
| IMS-China International Conference on Statistics and Probability                        | July 2024 |
| ICSA 2024 China Conference                                                              | June 2024 |
| School of Statistics and Data Science, Nankai University                                | May 2024  |
| School of Management, Fudan University                                                  | Apr 2024  |

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| School of Statistics and Management, Shanghai University of Finance and Economics       | Apr 2024  |
| The 2nd Chinese Causal Inference Conference                                             | Mar 2024  |
| School of Statistics & Institute of Statistics and Big Data, Renmin University of China | Nov 2023  |
| IMS New Researcher Conference                                                           | Aug 2023  |
| Joint Conference on Statistics and Data Science in China                                | July 2023 |
| ICSA 2023 China Conference                                                              | June 2023 |
| The 1st Chinese Causal Inference Conference                                             | May 2023  |
| School of Mathematics and Statistics, Wuhan University                                  | Apr 2023  |
| School of Data Science, The Chinese University of Hong Kong, Shenzhen                   | Mar 2023  |
| Interdisciplinary Seminar on “Data Science and Life and Health”                         | Oct 2022  |
| Department of Applied Mathematics, The Hong Kong Polytechnic University                 | Sep 2022  |
| The 5th International Conference on Econometrics and Statistics                         | June 2022 |
| Department of Family Medicine and Public Health, UC San Diego                           | Apr 2022  |
| ENAR 2022 Spring Meeting                                                                | Mar 2022  |
| 2021 Joint Statistical Meeting                                                          | Aug 2021  |
| Southern California Machine Learning Symposium (Poster), UC Los Angeles                 | Mar 2019  |
| Statistics & Data Science Symposium (Poster), UC San Diego                              | Jan 2019  |

## Service Activities

Council Member, Causal Inference Branch of the Chinese Association for Applied Statistics  
Council Member, Chinese Statistical Association of Young Scholars  
Council Member, Machine Learning Branch of the Chinese Association for Applied Statistics  
Council Member, Bayesian Statistics Branch of the Chinese Association for Applied Statistics

## Teaching

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Instructor, Institute of Statistics and Big Data, Renmin University of China                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2022 – present                                                                                                    |
| <ul style="list-style-type: none"> <li>• Statistical Models and Inference II (PhD level)</li> <li>• Probability Theory and Mathematical Statistics C (undergrad level)</li> <li>• Machine Learning (PhD level)</li> <li>• Causal Inference (PhD level)</li> <li>• Special Topics in Mathematical Statistics (PhD level)</li> <li>• Special Topics in Business and Economic Statistics (PhD level)</li> <li>• Special Topics in Biostatistics (PhD level)</li> <li>• Advanced Statistical Computing (PhD level)</li> </ul> | Spring 2026<br>Fall 2025<br>Fall 2025<br>Fall 2024<br>Spring 2023 – 2025<br>Fall 2023<br>Spring 2023<br>Fall 2022 |
| Teaching Assistant, Department of Mathematical Sciences, UC San Diego                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2018 – 2022                                                                                                       |

- Math 281A, 281B – Mathematical Statistics (PhD level)
- Math 189 – Data Analysis and Inference (undergrad level)
- Math 183 – Statistical Methods (undergrad level)
- Math 180A – Introduction to Probability (undergrad level)
- Math 11 – Calculus-Based Prob & Stats (undergrad level)